Ali Husain

AI SYSTEMS · INFRASTRUCTURE · WEB PRODUCTS

I design and build complete systems: *the interface people use, the data and services behind it, and the infrastructure that keeps it running.* Each one is specified first, tested before it ships, and documented to run without me.

1,257	15	9	2
automated tests, run in CI on every change	services run in production, 24/7	open-source projects, each documented and released	live offline-first apps, no account and no server

AREAS OF EXPERTISE

AI & LLM engineering
Agents, multi-agent workflows, LLM evaluation, prompt engineering, local models

Cloud, DevOps & SRE
Linux, containers, CI/CD, infrastructure as code, scheduled automation, monitoring, incident response

Networking & security
DNS, TLS, mesh VPN, firewalls, intrusion prevention, encryption, secrets handling

Backup & disaster recovery
Verified backups, multi-site copies, automated restore drills, rebuild runbooks

Software & web products
Python, TypeScript, Go, C/C++, Bash, SQL: offline-first web apps, services, command-line tools, technical writing

Generative media
Prompt systems and frame QA for image, video and music models; editing, sound design, loudness-checked 4K delivery

EXPERIENCE

AI Systems Engineer

`github / evalbench · aih · agent-guardrails`

Independent · agents, evaluation and model deployment

- **Built Evalbench, a self-hosted LLM evaluation workbench:** blind pairwise judging, seven automatic check types including code tests in a sandbox with no network, and regression tracking across local and API models; 107 automated tests.
- **Designed and run an autonomous AI operations agent** that monitors and maintains production infrastructure around the clock: bound by a written charter, woken by alerts, every change logged.
- **Built multi-agent delivery pipelines** of up to 9 coordinated agents (1.45M tokens, 566 tool calls) with build, critique and automated browser-verification stages, and ran 13-agent system audits.
- **Built a knowledge pipeline and agent guardrails:** 4 GB+ of conversations from three machines in one full-text archive, LLM extraction cross-checked by a second model, and a hook that screens agent commands before they run.

Infrastructure & DevOps Engineer

`github / homelab · cockpit`

Independent · self-hosted production platform, run as code

- **Operate a 15-service containerized platform** (14 health-checked) behind one TLS edge: DNS, password and identity management, file sync, smart-home control; remote access over a 7-node mesh VPN with zero port forwards.
- **Built Cockpit, a read-only operations dashboard:** one 8 MB Go binary with a TypeScript UI fed by Server-Sent Events, run as privilege-split systemd services; 113 automated tests, including a 33-check end-to-end suite under real systemd.
- **Designed backup & disaster recovery:** 12 integrity-verified sets nightly, 3 copies on 3 machines, automated weekly restore drills, a 2–3 hour bare-metal rebuild plan.
- **Built observability for 77 failure conditions** plus 13 uptime probes at 60 s, self-healing watchdogs and a dead-man switch; everything as code: 376 commits, 20 scheduled services, drift detection, ~3,800 lines of runbooks and notes.

CONTACT

EMAIL
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PORTFOLIO
gallowforge.github.io

GITHUB
github.com/gallowforge

LOCATION
Remote · UTC+3

TOOLBOX

AI platforms
Anthropic and OpenAI models, open-weight LLMs, local inference on GPU, agent frameworks, MCP, Hugging Face ecosystem

Generative media
Image, video, voice and music generation models; diffusion pipelines; programmatic editing with Python and ffmpeg

Infrastructure
Linux (Debian, Arch), Docker, systemd, Git, GitHub Actions, Caddy, Nginx, Tailscale / WireGuard, SQLite, Redis, CouchDB

Web & product
Astro, React, Preact, Tailwind, FastAPI, PWAs, Playwright, Vitest, Lighthouse CI, accessibility (WCAG)

Languages
Python, TypeScript / JavaScript, Go, C, C++, Bash, SQL, R, Standard ML, Prolog

OPEN TO

- **Remote contracts:** web products, infrastructure, DevOps, automation, AI integration, video and audio post-production.
- **AI training & evaluation:** code review, model-output rating, prompt and rubric writing.
- **Flexible hours:** UTC+3 overlaps the full European working day.

Web & Product Developer

github / Ethereal · ledger

Independent · product design, front-end and delivery

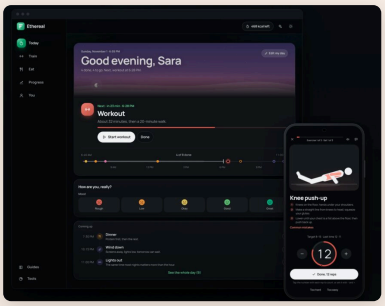
Ethereal

LIVE

Offline-first health & fitness app · product lead

Led product design and delivery of an installable, account-free app that plans the day, coaches workouts and tracks nutrition. 41 animated exercises, a 1,474-food offline database and 392 automated tests, built to WCAG 2.1 AA and Lighthouse 100. gallowforge.github.io/Ethereal

TypeScript · Astro · PWA · CI/CD · UX design



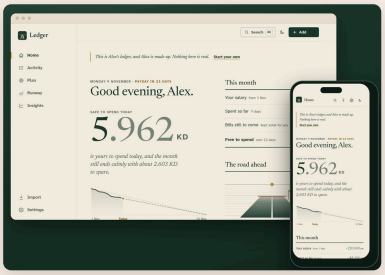
Ledger

LIVE

Offline-first personal finance app · product lead

Turns savings into a monthly salary that lasts: what is safe to spend today and how long the money runs. Reads bank statements on the device and reconciles every row against the running balance. No account, no server; 483 automated tests. gallowforge.github.io/ledger

TypeScript · Astro · Preact · pdf.js · PWA



Sites, 3D and 40+ interactive tools

A six-page animated brand site for an AI video-ad studio; a scroll-driven WebGL story page verified by automated screenshots at phone and desktop sizes; live market dashboards on real-time price feeds, learning tools, data visualizations and planners, each designed for phone and desktop.

Next.js · Tailwind · Framer Motion · WebGL · Puppeteer · real-time data

Video, Audio & Generative Media Producer

github / filmforge

Independent · production method, editing, sound and delivery

- **Built filmforge, a documented production method** for painted films made with AI image and video models: nine Claude Code skills (story, style lock, prompt grammar, frame QA, motion, sound, delivery) and seven tested Python tools that price, lint and verify the work at every stage gate.
- **Post-production built in code:** beat-matched editing, color grading, motion graphics and animated titles, narration timed across cuts, layered sound design and mixing to loudness targets, 4K mastering on GPU.
- **Delivery checked, not assumed:** every encode verified for streams, loudness and picture-to-music sync, and each production archived with a SHA-256 completeness proof anyone can re-verify.

Open-Source Tools

github / agent-guardrails · aih · bwtidy

Independent · small tools, each tested and released

- **agent-guardrails:** reviewer and verifier agents that judge work they did not write, and a hook that warns a coding agent before a destructive shell command runs; 36 automated tests.
- **aih:** every AI conversation from three machines in one local full-text index, searchable from the command line; 31 automated checks.
- **bwtidy:** audits and tidies a password vault entirely on the device, and changes nothing until you approve the plan; 12 automated tests.

TESTED IN PUBLIC

ledger	483
Ethereal	392
cockpit	113
evalbench	107
homelab	43
filmforge	40
agent-guardrails	36
aih	31
bwtidy	12

Automated tests per repository, run in CI on every change: 1,257 in all.

HOW I WORK

- **Prove it works.** A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.
- **Write it down.** Every system ships with a runbook someone else could follow.
- **Fewer things, done properly.** Simple setups that stay up beat clever ones that break.
- **Learn by shipping.** Linux, Go, AI agents, post-production: each learned by building something real.

EDUCATION

Computer Science and Chemical Engineering, Kuwait University

Continuous self-study

Algorithms, computer science and AI, each put to use in a shipped project

BEYOND WORK

Cinema (1,150+ films logged) and storytelling.

Scan for code and live work